

1. Administrative & Study Identification

Study Title A Study to Determine the Efficacy and Safety of Deucravacitinib Compared With Placebo in Participants With Active Psoriatic Arthritis (PsA) Who Are Naïve to Biologic Disease-modifying Anti-rheumatic Drugs **Official Title** A Phase 3, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Efficacy and Safety of Deucravacitinib in Participants With Active Psoriatic Arthritis Who Are Naïve to Biologic Disease-modifying Anti-rheumatic Drugs **Short Title/Acronym** POETKY PsA-1 **Protocol Number** IM011054 (IM011-054) **NCT Number** NCT04908202 **Posting ID** 608279908339159511 **Sponsor Name / Company** Bristol Myers Squibb **Investigational Product** Deucravacitinib (Trade Name: Sotyktu) **ATC Code** L04AA56 **Phase of Development** PHASE3 **Trial Status** ACTIVE_NOT_RECRUITING **Medical Monitor** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy and safety of deucravacitinib versus placebo in improving joint and skin symptoms in participants with active psoriatic arthritis (PsA) who are naïve to biologic disease-modifying anti-rheumatic drugs. * **Secondary Objectives:** To evaluate the effect of deucravacitinib on PASI 75 (Psoriasis Area and Severity Index), Minimal Disease Activity (MDA), resolution of enthesitis and dactylitis, and improvements in patient-reported outcomes including HAQ-DI, SF-36 PCS, and FACIT-Fatigue.

2.2 Background & Rationale Psoriatic arthritis is a complex, multifaceted, and heterogeneous autoimmune disease that can severely impact joint and skin health. While biologic therapies exist, there remains a significant unmet need to equip health care providers with safe and effective oral treatment options. Deucravacitinib is a first-in-class oral, selective, allosteric tyrosine kinase 2 (TYK2) inhibitor. This study is conducted to demonstrate its superiority over placebo in biologic-naïve adult patients, aiming for durable improvements in joint/skin symptoms and overall quality of life.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled * **Blinding:** Double-blind * **Randomization:** 1:1

3.2 Planned Enrollment & Duration * **Target Enrollment:** 670 participants * **Number of Sites:** Global multicenter trial * **Estimated Study Start:** 13-Jul-2021 * **Estimated Primary Completion:** 05-Sep-2024

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria** 1. Diagnosed to have psoriatic arthritis (PsA) of at least 3 months duration at screening and meets the Classification Criteria for Psoriatic Arthritis. 2. Active arthritis as shown by ≥ 3 swollen joints and ≥ 3 tender joints at screening and Day 1, with high sensitivity C-reactive protein (hsCRP) ≥ 3 mg/L. 3. Active plaque psoriatic skin lesion(s) or documented medical history of plaque psoriasis (PsO), and ≥ 1 PsA-related hand and/or foot joint erosion on X-ray confirmed by central reading.
- 4.2 Exclusion Criteria** 1. Received an approved or investigational biologic therapy for the treatment of PsA or PsO. 2. Nonplaque psoriasis at screening or Day 1. 3. History of or current inflammatory joint disease other than PsA (e.g., gout, reactive arthritis, rheumatoid arthritis) or other autoimmune conditions such as systemic lupus erythematosus or active fibromyalgia.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen** * **Dosage/Frequency:** Deucravacitinib 6 mg once daily (QD). * **Route of Administration:** Oral * **Duration of Treatment:** 52 weeks (followed by an optional open-label long-term extension period).
- 5.2 Concomitant & Prohibited Therapy** * **Permitted:** Standard non-biologic DMARDs and background standard-of-care permitted per protocol. * **Prohibited:** Biologic disease-modifying anti-rheumatic drugs (bDMARDs) and other investigational therapies.
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, X-Rays, Joint Assessment, hsCRP Labs
Baseline	Day 0	Randomization, First Dose, Baseline PROs and Clinical Scores	Interim Week 16 Efficacy Assessment (Primary Endpoint - ACR20), Safety Labs, PRO collection
End of Study	Week 52	Final 52-Week Efficacy Assessment, Entry into Long-Term Extension Period	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Proportion of participants achieving an ACR20 (American College of Rheumatology 20% improvement) response at Week 16. * **Measurement Method:** Clinical joint assessment (tender/swollen joint counts) and patient/physician global assessments.

7.2 Secondary & Exploratory Endpoints * Proportion of participants achieving PASI 75 and Minimal Disease Activity (MDA). * Resolution of enthesitis and dactylitis (using the SPARCC index). * Change from baseline in HAQ-DI, SF-36 PCS, FACIT-Fatigue, mSvdH total score, and DAS28-CRP.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous collection via clinical evaluation, laboratory tests, and patient reports at each study visit. All non-serious AEs (including SARS-CoV-2 related AEs) collected from the initiation of study treatment.
 - **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness and followed until resolution or stabilization.
 - **Data Monitoring Committee (DMC):** Independent safety monitoring board evaluating ongoing blinded safety data throughout the trial phase.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Set for all participants who received at least one dose of study medication.
 - **Sample Size Justification:** Target Enrollment of 670 randomized 1:1 to provide adequate statistical power ($\alpha = 0.05$) to detect superiority of deucravacitinib over placebo in ACR20 response.
 - **Handling of Missing Data:** Non-responder imputation (NRI) for binary categorical endpoints (e.g., ACR20, PASI 75); Mixed Model for Repeated Measures (MMRM) for continuous secondary endpoints.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

- **Monitoring Plan:** Routine onsite and remote monitoring to verify protocol compliance, patient safety, and accuracy of data capture.
 - **Audit Trail:** Validated electronic data capture (EDC) system with full audit trail capabilities; centralized reading for X-ray assessments to ensure standardization.
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11. Study Identifiers & Governance

Primary ID: IM011054 **Registry Number:** NCT04908202 **Posting ID:** 608279908339159511 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Determine the Efficacy and Safety of Deucravacitinib Compared With Placebo in Participants With Active Psoriatic Arthritis (PsA) Who Are Naïve to Biologic Disease-modifying Anti-rheumatic Drugs **Official Regulatory Title:** A Phase 3, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Efficacy and Safety of Deucravacitinib in Participants With Active Psoriatic Arthritis Who Are Naïve to Biologic Disease-modifying Anti-rheumatic Drugs **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/sotyktu-epar-product-information_en.pdf **Source Level:** 5

12. Clinical Framework (Psoriatic Arthritis Specific)

Primary Condition / Disease Areas: Psoriatic Arthritis **Preferred UMLS Name:** Psoriatic Arthritis **Diagnosis Threshold:** ≥ 3 swollen joints, ≥ 3 tender joints, and hsCRP ≥ 3 mg/L **Medical Methodology:** Selective TYK2 Inhibitor Oral Therapy **Optimization Target:** ACR20 response and Minimal Disease Activity (MDA)

13. Study Procedures & Methodology

Management Pathway: Oral systemic targeted therapy for biologic-naïve active PsA **Education Protocol (HCP):** Protocol training, standardized joint count assessment training, and central X-ray procedure alignment **Education Protocol (Patient):** PRO completion instructions and study medication administration guidance **Quality Audit Mechanism:** Centralized radiographic reading for joint erosions and standard regular EDC monitoring

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (followed by optional long-term extension) **Onsite Visit Cadence:** Screening, Baseline, Week 16, Week 52, and intermediate scheduled safety assessments **Remote Monitoring Frequency:** Routine data/AE monitoring according to

trial schedule **Checklist Review Interval:** Continuous safety follow-up up to 30 days after the last dose

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				